Kali McDougall, MSc.

University of Victoria
ICE Remote Sensing Lab

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LinkedIn Profile

Education

University of Victoria *Sept 2022 - Apr 2025*

Master of Science - Glaciology & Remote Sensing

- Thesis title: Optimization of C- and L-band synthetic aperture radar for all-season rift detection on the Larsen C Ice Shelf, Antarctica
- Supervisor: Dr. Randall Scharien

University of Victoria *Sept 2019 - Apr 2022*

Honours Bachelor of Science - Physical Geography & Geomatics

- Thesis title: Synoptic drivers of ice-jam flooding in the Peace-Athabasca Delta, AB
- Supervisor: Dr. David Atkinson

Awards & Distinctions

American Geophysical Union Student Travel Award, American Geophysical Union *Oct 2024*

Student Participation in Space Conferences Grant, Canadian Space Agency *Oct 2023 & 2024*

Northern Scientific Training Fund, POLAR Knowledge Canada *June 2023 & 2024*

Graduate Fellowship Award, University of Victoria *Sept 2022*

2021-2022 Honours in Geography, University of Victoria *Apr 2022*

Education Achievement Bursary, BC Youth Federation *May 2019*

Background & Research Interests

My academic background is concentrated in cryospheric science, climatology, and microwave remote sensing. My research interests broadly include the use of satellite remote sensing to study glaciers and ice sheets and their connection to the polar and global climate systems. I have experience using Python programming (GeoPandas, Rasterio, GDAL, xarray), machine learning, and a wide range of geomatics applications (QGIS, ArcGIS, ESA SNAP, Catalyst Earth, Google Earth Engine) to process and analyze remote sensing data types such as SAR, LiDAR, optical, and passive microwave.

Research Experience

Sikuttiaq: Sea ice travel safety and monitoring in Inuit Nunangat *June 2023 - Apr 2026*

Canada-Inuit Nunangat-United Kingdom (CINUK) Research Programme, Pond Inlet, NU

- Sea ice roughness, thickness, and slush are key characteristics that not only determine the safety and efficiency of ice travel for Inuit communities, but also hold significant implications for their overall well-being. This work was carried out as part of the Sikuttiaq project, which aims to identify and map sea ice hazards to promote safe ice travel for Inuit. This work involved the application of a regression tree sea ice roughness model developed using C-band SAR, and validation of the model using coincident UAV- and ALS-derived DEMs. Additionally, this work involved a field campaign during which a GEM-2 electromagnetic sensor and manual sampling techniques were used to characterize slush and seawater flood zones on melting sea ice. These efforts laid the groundwork for future field campaigns and produced ground truth data for the Sikuttiaq project.

MSc Thesis: Optimization of C- and L-band synthetic aperture radar for all-season rift detection on the Larsen C Ice Shelf, Antarctica *Sept 2022 - Apr 2025*

University of Victoria, Victoria, BC

- More than half of ice mass loss from Antarctica occurs through iceberg calving at the outer margins of floating ice shelves. These events are associated with the propagation of large rifts which extend through the entire thickness of the ice, and this process has been identified as a precursor to major disintegration events. SAR provides the best utility for continuous monitoring of rifts, though the constraints on detection posed by SAR imaging parameters must be characterized. This work focused on optimizing SAR for ice shelf rift detection using Sentinel-1 C- and PALSAR-2 L-band SAR applied to automated detection techniques. Rift detectability was assessed as a function of SAR frequency and polarization, ASCAT derived surface melt conditions, and ICESat-2 derived measurements of rift geometry. Lastly, pixel- and object-based Random Forest classification algorithms were compared to determine the most appropriate technique for classifying different ice and rift types over the Larsen C Ice Shelf.

Image processing-based atmospheric river detection over the Antarctic Peninsula *Sept - Dec 2022*

University of Victoria, Victoria, BC

– Landfalling atmospheric rivers contribute large amounts of snowfall and cause strong melt events across the Antarctic Ice Sheet, which have been shown to catalyze large iceberg calving events. Most methods of detecting atmospheric rivers rely on magnitude thresholding, which may pose challenges in the future due to increasing atmospheric moisture. This project evaluated the performance of an image-processing based technique to detect atmospheric rivers irrespective of their magnitude. A top-hat by reconstruction algorithm based on ERA5 integrated vapour transport data was used to detect and track a series of atmospheric rivers that preceded major calving events on the Antarctic Peninsula. The algorithm successfully identified atmospheric rivers at a variety of magnitudes and spatial scales with reduced sensitivity to detection parameters. This suggests that the top-hat by reconstruction algorithm overcomes limitations posed by magnitude-dependent detection techniques and may be more appropriate for tracking atmospheric rivers under a warming climate.

BSc Thesis: Synoptic drivers of ice-jam flooding in the Peace-Athabasca Delta, AB *Sept 2021 - Apr 2022*

University of Victoria, Victoria, BC

– The Peace-Athabasca Delta is currently threatened by decreasing frequency of high-magnitude ice-jam floods, which are necessary to restore water to ecologically sensitive elevated basins. To understand the cause behind ice-jam floods and how they are affected by a changing climate, it is necessary to study broad-scale synoptic patterns which provide an overview of basin-wide hydrologic activity. Linux command line operators and Python geospatial packages were applied to 700 mb geopotential height composites to compute seasonal averages and reveal the dominant synoptic patterns behind major flood years in the delta. Results showed that major flood years are associated with atmospheric ridging over the northeast Pacific which promotes a poleward shift of the jet stream and enhances Arctic air advection and cyclonic activity over north-central Canada. This results in a larger snowpack in the headwaters and a thicker river ice cover, thus increasing the probability of mechanical ice-jamming and flooding.

Variability of supraglacial lake extent and ice velocity on the Kangerlussuaq Glacier *Jan - Apr 2022*

University of Victoria, Victoria, BC

– Mass balance of tidewater glaciers in Greenland is the product of a complex set of atmospheric, oceanic, and geometric glacial interactions that operate on time scales ranging from daily to decadal. This project examined trends in supraglacial lake extent, terminus position, synoptic climate, and ice velocity near the terminus of the Kangerlussuaq Glacier from May 2018 to December 2021, in order to estimate the effect of surface melt on glacier sliding and ice mass loss. Sentinel-1 SAR imagery was used to derive terminus position and ice velocity using the offset tracking technique for monitoring ice motion. Trends were compared to 700 mb geopotential height composites and delineations of supraglacial lake extent produced from the NDWIce band ratio applied to Sentinel-2 optical imagery. Results showed that growth in supraglacial lake extent precedes ice velocity increases and periods of terminus advance, while synoptic conditions related to the surface mass balance of the glacier explain limited observed variance in velocity. This implies that the Kangerlussuaq Glacier remains in a phase where terminus geometry and ice dynamics are the dominant control of ice mass loss, at least on interannual timescales.

The role of radiation in surface melt and sediment transport on the Leverett Glacier *Jan - Apr 2021*

University of Victoria, Victoria, BC

– The mass balance of ice masses is largely determined by their surface energy balance, which drives the evacuation of meltwater and the distribution of sediment in glacial outlet streams. Understanding the effects of radiation on mass balance is therefore necessary to model downstream landscape change. The energy balance and mass balance of Greenland's Leverett Glacier were modeled using in situ ice temperature and radiation data. ASMR-E passive microwave data were used to examine the seasonal progression of surface melt and 2m DEMs from 2010 and 2012 were used to investigate proglacial elevation change as a proxy for sediment transport. The melt model revealed that almost 100% of incoming radiation is reflected by the ice surface at the beginning of the melt season, with this percentage decreasing throughout the summer as energy is absorbed by the ice and put towards melt. This correlates well with increased sediment deposition along the proglacial outlet stream and a reduction in ice sheet elevation during the summer. This suggests that downstream landscape change is strongly dependent on the surface mass balance of the glacier which drives seasonal meltwater evacuation and proglacial sediment transport.

Professional Experience

Remote Sensing Specialist *Apr 2024 - Apr 2026*

SmartICE Sea Ice Monitoring and Information, Victoria, BC

– Applied a regression tree model to C-band SAR imagery from the RADARSAT Constellation Mission to identify regions of significant sea ice roughness throughout the Canadian Arctic. Compared model results against roughness retrieved from UAV- and ALS-derived DEMs and LiDAR data. Used the results to create detailed hazard maps aimed at enhancing ice safety and navigation for Inuit communities.

– Participated in a field campaign in Pond Inlet, Nunavut, which sought to develop protocols to characterize slush and seawater flood zones in Eclipse Sound using electromagnetic sensors and manual sampling techniques.

Research Assistant *May - Aug 2022*

University of Victoria, Victoria, BC

- Developed Python code to identify and characterize anomalous atmospheric circulation in northwestern Canada, focusing on the implications for ice-jam flooding in the Peace-Athabasca Delta. Created algorithms to process large climate datasets and highlight unusual weather events and their potential impacts.
- Summarized results and produced scientific literature for Environment and Climate Change Canada outlining the geophysical and climatological controls on ice-jam flooding in the delta.

Teaching Experience

Specialist Instructor *Jan 2023 - Dec 2025*

University of Victoria, Victoria, BC

GEOG 228: Intro to Remote Sensing

- Led laboratory exercises and provided support to students during tutorials and exams. Labs covered fundamental remote sensing concepts and included exercises on interpreting airborne and spaceborne remote sensing products, performing geometric, radiometric, and atmospheric corrections, image enhancements and transformations, and classifications using Catalyst Earth and Google Earth Engine.

GEOG 422: Advanced Topics in Remote Sensing

- Guided students' capstone projects and led exercises designed to help students develop fundamental research skills. Lab topics covered advanced remote sensing techniques and included the use of cloud-computing to process and analyze satellite imagery. Developed a Python-based melt onset detection lab using SAR imagery and resources from the Alaska Satellite Facility.

GEOG 370: Advanced Hydrology

- Led laboratory exercises and provided support to students during tutorials and exams. Lab topics covered advanced atmospheric, groundwater, and surface hydrology concepts and exercises included the use of QGIS to perform watershed mapping and hydrologic pattern analysis. Coordinated field-based assignments and chaperoned off-campus demonstrations.

Volunteer Experience

Polar EO Project Group Member *Oct 2023 - June 2025*

Association of Polar Early Career Scientists (APECS)

- Developed a Github database for polar remote sensing resources as part of APECS' Polar Earth Observation Database project group. Assisted with web development and content creation, and connection with Github database.

Co-Director of Events *Sept 2023 - Apr 2024*

UVic Women in Science

- Coordinated educational and social events aimed at supporting women in STEM fields as part of the Events Committee. Hosted the annual Research Symposium to showcase women-led research at UVic.

Graduate Student Representative *Sept 2022 - Aug 2023*

UVic Graduate Student Society

- Represented Geography graduate students in major governance decisions on university policy as a part of the Graduate Representative Council and the Student Affairs Committee.

Conference Presentations & Talks

Advancing Ice Shelf Rift Detection with Synthetic Aperture Radar: Comparing C- and L-band Frequency over the Larsen C Ice Shelf (invited) *Dec 2024*

NASA Goddard Institute for Space Studies, New York, NY

Larsen C Ice Shelf rift detection and evolution using C- and L-band synthetic aperture radar *Dec 2024*

American Geophysical Union 2024 Annual Meeting, Washington, DC

Larsen C Ice Shelf rift detection and evolution using C- and L-band synthetic aperture radar *Sept 2024*

2024 European Polar Science Week, Copenhagen, DK

Detection of rifts on the Larsen C Ice Shelf using C- and L-band synthetic aperture radar *Dec 2023*

American Geophysical Union 2023 Fall Meeting, San Francisco, CA

Detection of rifts on the Larsen C Ice Shelf using C- and L-band synthetic aperture radar *Oct 2023*

9th Association of Polar Early Career Scientists International Online Conference, Tromsø, NO (virtual)

Synoptic drivers of ice-jam flooding in the Peace-Athabasca Delta *Mar 2022*

Circumpolar Students' Association Northern Research Day 2022, Edmonton, AB (virtual)